OFFICE OF UNDERGRADUATE RESEARCH
IMMERSIVE RESEARCH INTERNSHIP EXPERIENCE

Colorimetric Assay For MSUD Detection Through Alpha Keto Acids

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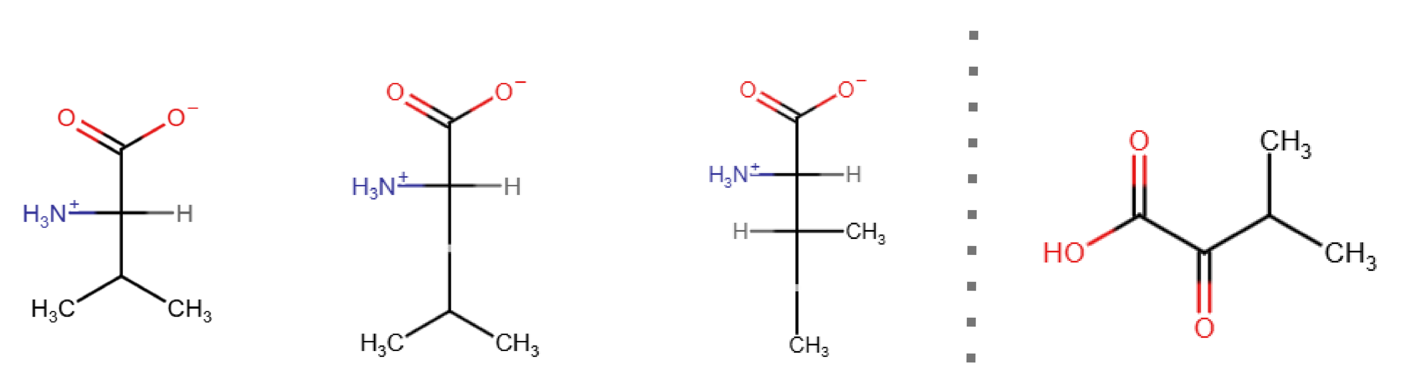
Office of Undergraduate Research, University of Maryland

FIRE: THE FIRST-YEAR INNOVATION & RESEARCH EXPERIENCE
RAPID DIAGNOSTICS

Introduction

Maple syrup urine disease (MSUD) is a rare genetic disorder affecting the metabolism of branched-chain amino acids, causing the buildup of toxic byproducts, branched-chain α-ketoacids, in the body.



Left to right: three branched-chain amino acids (valine, leucine, and isoleucine), and byproduct of valine, α-ketoisovalerate

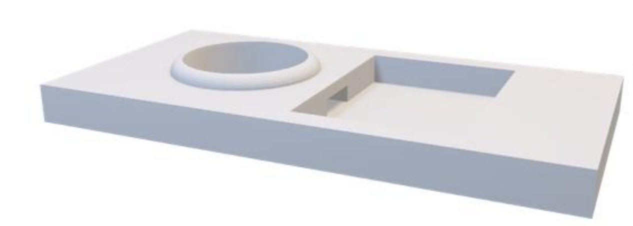
Symptoms of classical MSUD, the most severe form, include: delayed developmental milestones, lethargy, difficulty feeding, and sweet-smelling urine.
Left untreated, it leads to irreversible neurological injury within 7 to 10 days after birth, and death occurs within 2 months[1].

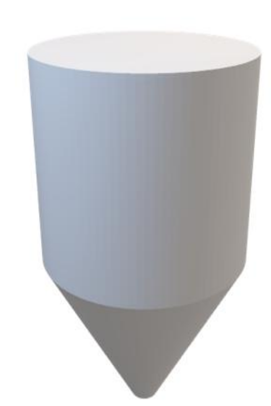
Testing for MSUD consists of a blood or urine screening, which takes time and advanced technology. Our goal is to develop a simpler colorimetric test for the disease, targeting a byproduct of valine, α-ketoisovalerate (KIV).

Device Concept

Vial with DNPH and pH of ~2 (regulated by Sulfuric Acid), and designed to attach to test box

Test box with a port for the vial with the sample and a test area that contains a strip that is prepped with ninhydrin and enough sodium hydroxide to create a pH of ~13



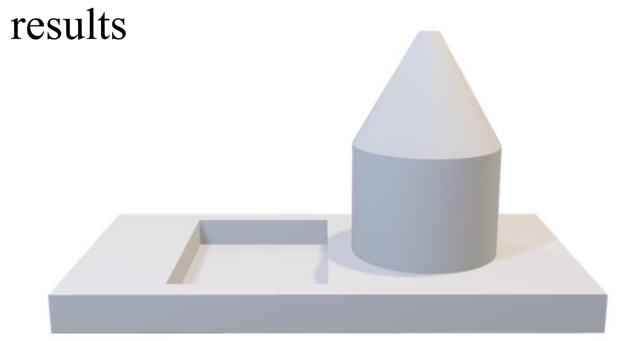


I. Add urine sample to vial

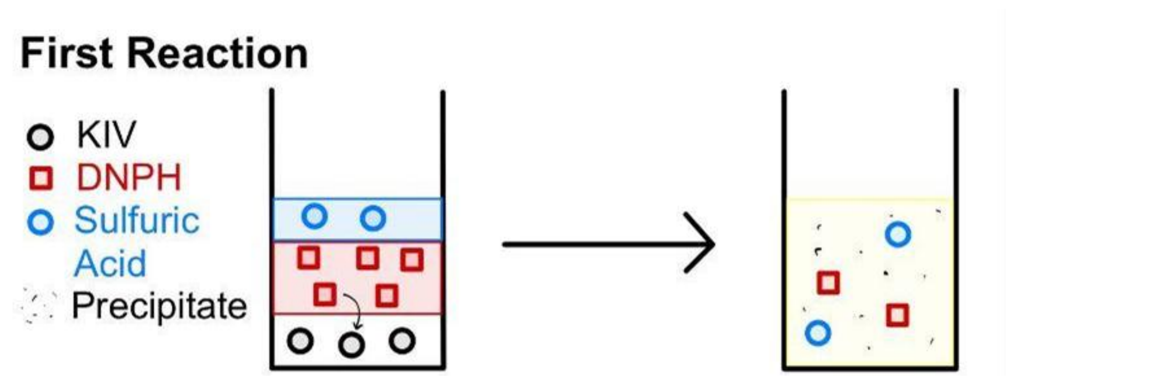
II. Shake well for 5 seconds

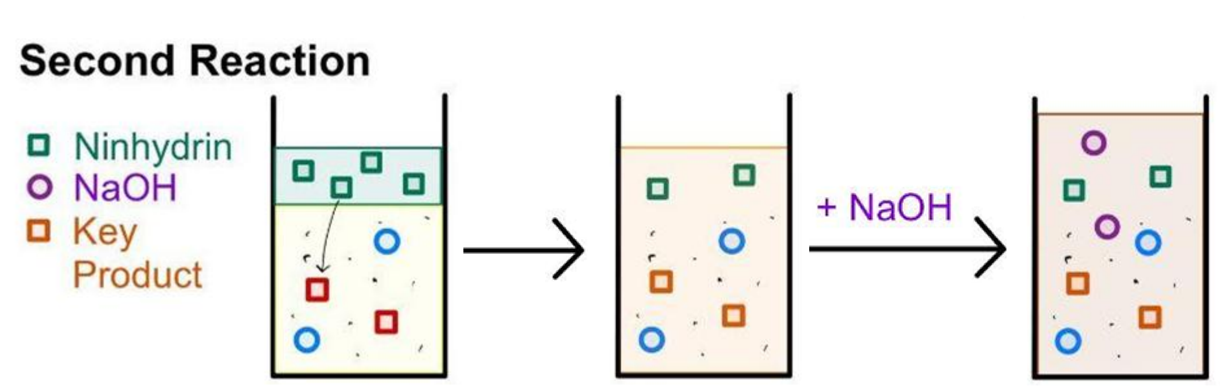
III. Let rest for 10 minutes

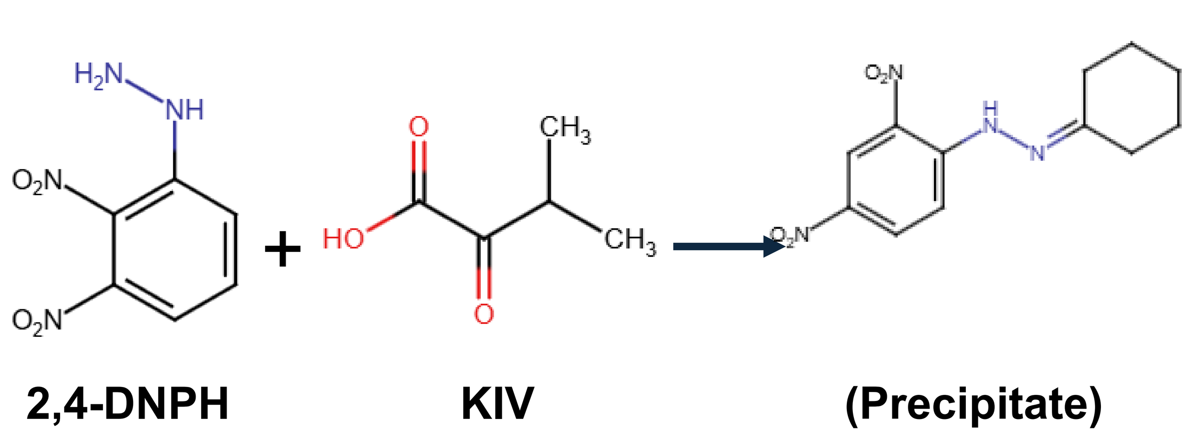
Attach vial to port on test box and compare results with legend on left hand side of the test box to determine positive or negative results

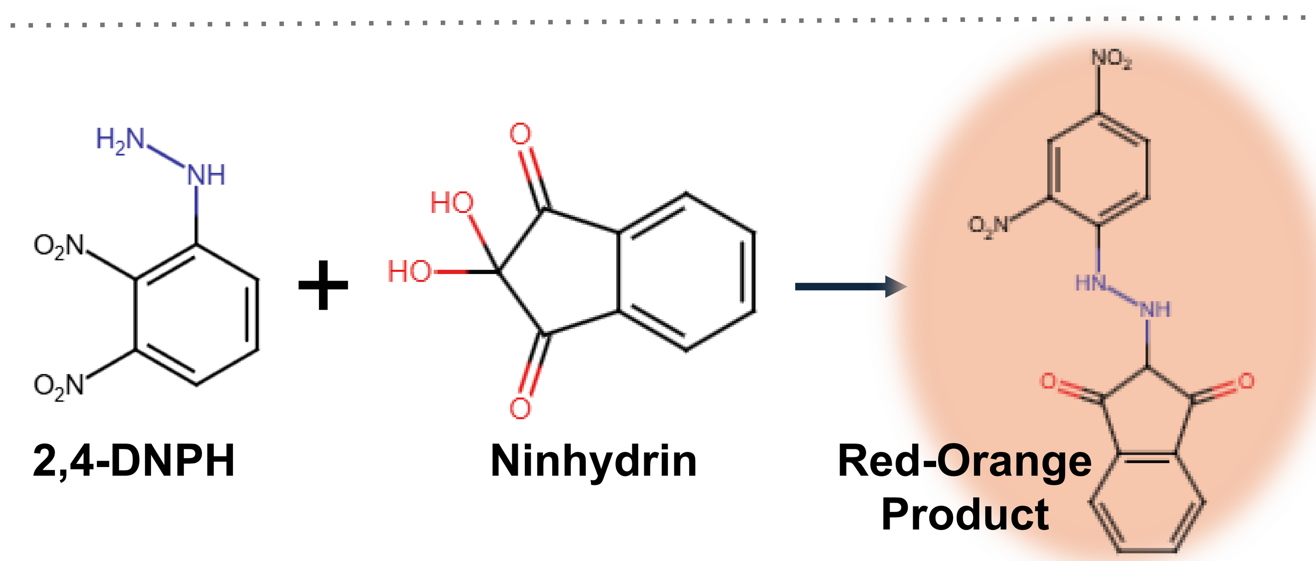


Methods



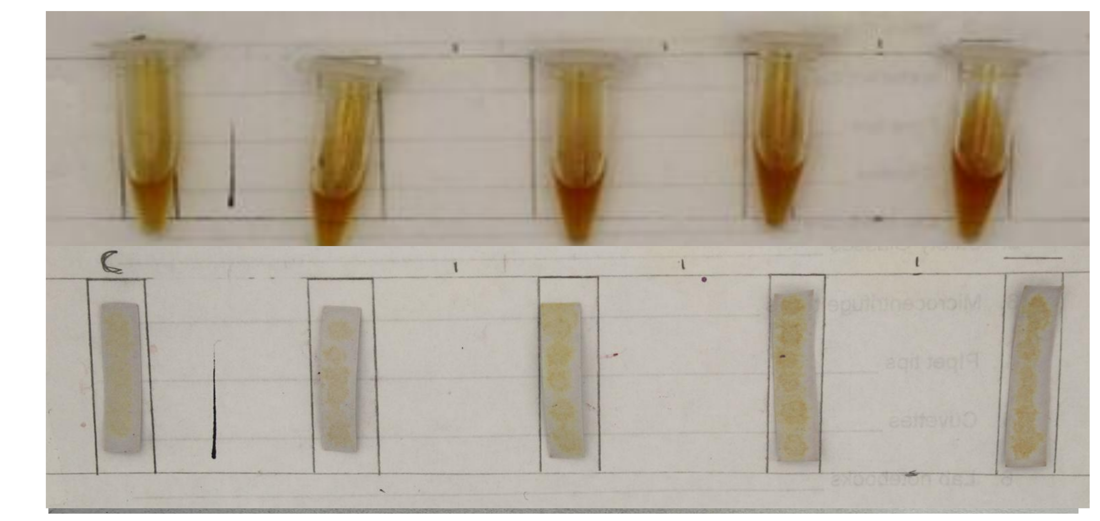




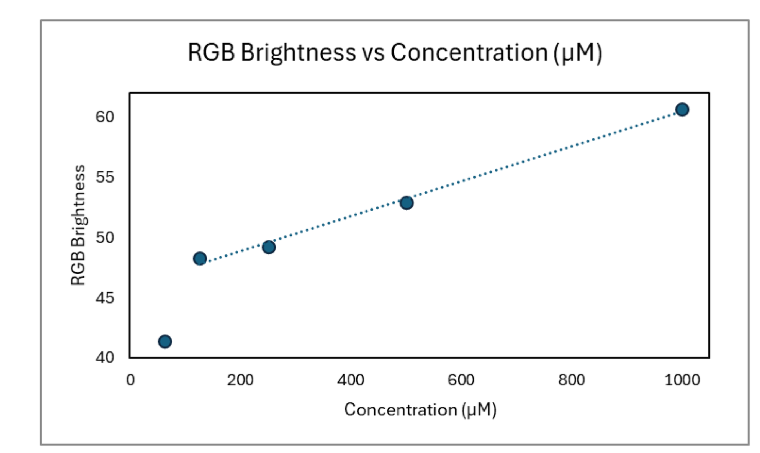


Our method involves two reactions. The target, KIV, reacts with 2,4-dinitrophenylhydrazine (2,4-DNPH), forming an imperceptible precipitate. Remaining 2,4-DNPH then reacts with ninhydrin, forming a red-orange color. Lower KIV results in less 2,4-DNPH used in the first reaction, allowing for a more intense color change. Higher concentrations of KIV result in more 2,4-DNPH used, leading to less red-orange color formed.

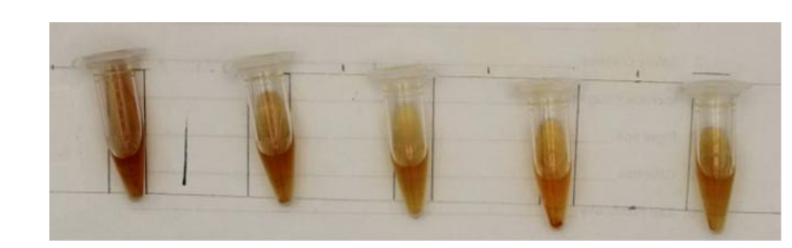
Results/Discussion



Reactions of 1:2 serial dilutions of KIV in solution starting at 1 mM KIV (left), 1 mM DNPH, 0.1 M sulfuric acid, 8% ninhydrin, and 1 M NaOH. This reflects our current level of sensitivity, where 0.5 mM and higher concentrations would result in a positive test and 0.25 mM and lower concentrations would result in a negative test. The liquid tests have proven to be much more differentiable between different concentrations compared to paper tests, which did not show as much differentiability.



Plot of the concentrations of KIV vs. RGB brightness values from our final liquid tests. Higher RGB values indicate more brightness (higher is more yellow, lower is more red). Trendline shown for the linear portion formed by the last four data points.



Original experiments, including 5 mM KIV (left), 5 mM DNPH, 2% ninhydrin, and 1 M NaOH, did not involve sulfuric acid, which was necessary to make Brady's Reagent rather than just 2,4-DNPH dissolved in ethanol. The exclusion of sulfuric acid gave us a color gradient that was opposite to the expected results.

Next Steps

- Produce a more significant color change below 200 μM KIV (test of these lower concentrations with current methods shown in image)
- Develop a method to be able to show a color change (red) at a KIV concentration of 1 μM
- Further develop and finish a usable paper fluidic device

Serial dilutions of 50 μM DNPH (left) also containing 8% ninhydrin and 2 M NaOH.

References

1. Hassan SA, Gupta V. Maple Syrup Urine Disease. [Updated 2024 Mar 3]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK557773/
2. Elgendy, K.; Alaa, M.; Elmosallamy, E.; Hassan, M.; Gamal, H. Applications of Ninhydrin: Spectrophotometric Determination of Hydrazine and Hydroxylamine. *Malaysian Journal of Analytical Sciences* **2022**, 26, 855–866.
3. Structures drawn with Marvin JS by Chemaxon.

Acknowledgements

Thank you to the Immersive Research Internship Experience (IRIE) and the First Year Innovation and Research Experience (FIRE) programs at the University of Maryland.

FIRE: THE FIRST-YEAR INNOVATION & RESEARCH EXPERIENCE

RAPID DIAGNOSTICS

Colorimetric Paperfluidic Assay for MSUD Detection Through Alpha Keto Acids

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Introduction

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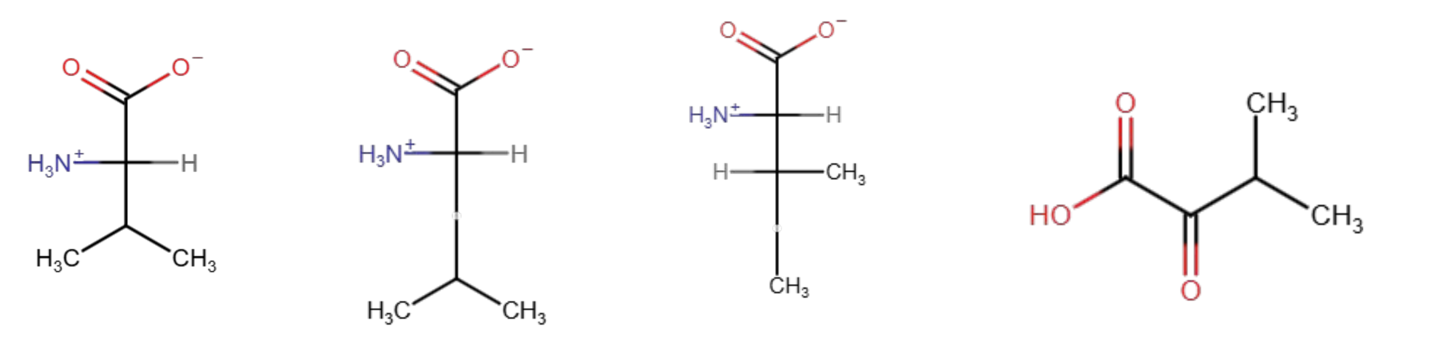
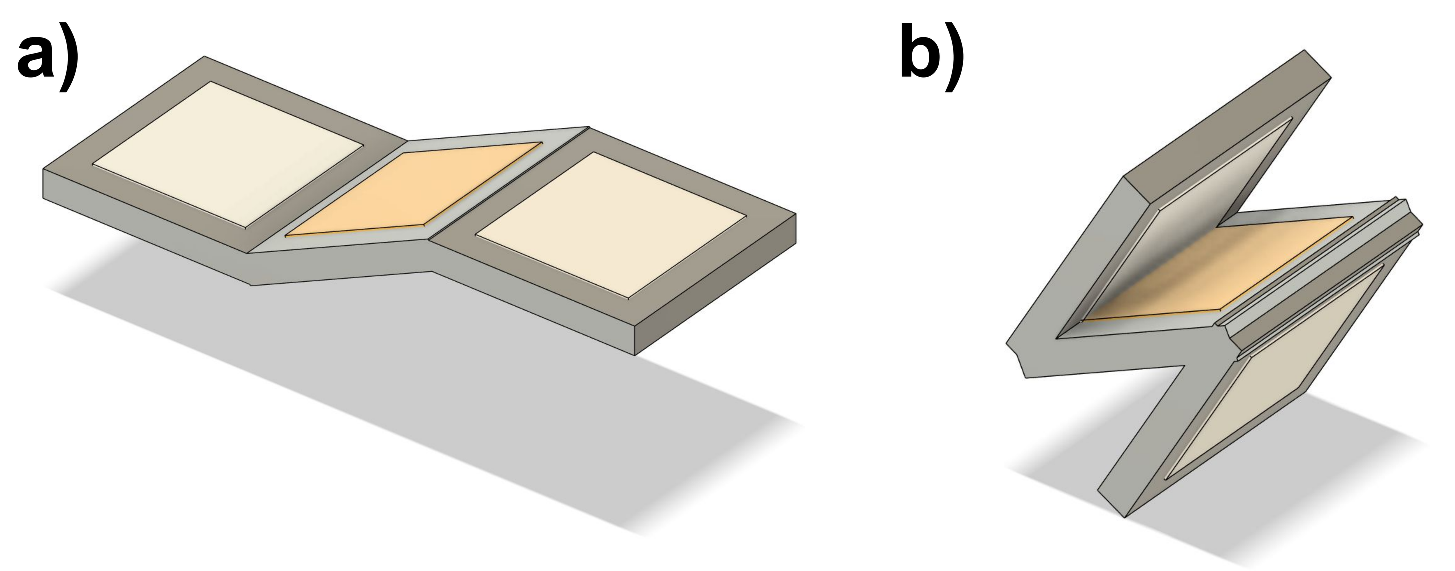


Left to right: 3 branched-chain amino acids valine, leucine, and isoleucine, and KIV

Symptoms of classical MSUD, the most severe form, include delayed developmental milestones, lethargy, difficulty feeding, and sweet-smelling urine. Left untreated, it leads to irreversible neurological injury within 7 to 10 days after birth, and death occurs within 2 months[1].

Testing for MSUD consists of a blood or urine screening, which takes time and advanced technology. Our goal is to develop a simpler paperfluidic colorimetric device for the disease, targeting a byproduct of valine, α-ketoisovalerate (KIV).

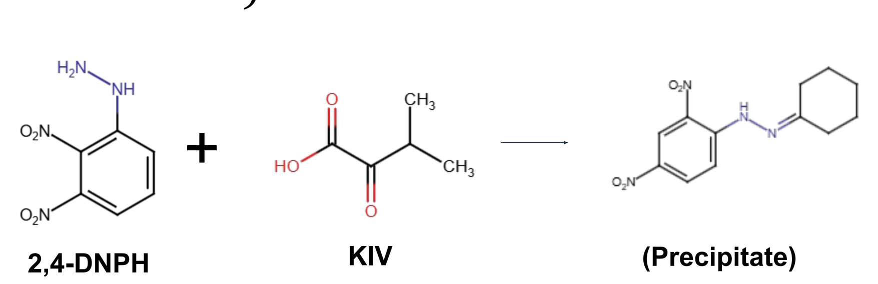
Device Concept



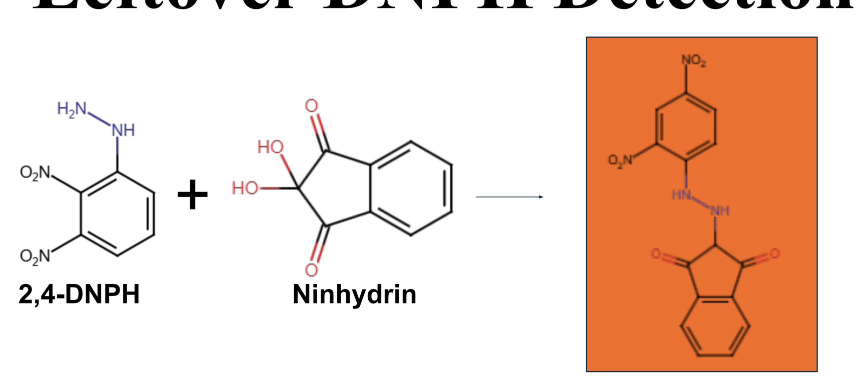
The origami paperfluidic device proposed has 3 paper spots for samples (shown on both sides for accessibility to middle). Following (a), the left spot contains sodium hydroxide, the middle contains Brady's reagent (DNPH and Sulfuric Acid Combined), and the right contains ninhydrin. KIV sample is added to the center spot, and then ~5-10 minutes should pass until the ninhydrin spot should be folded onto it. Lastly, the sodium hydroxide is folded onto the center spot from the opposite direction, as seen in (b). The device can then be heated to optimize results.

Methods

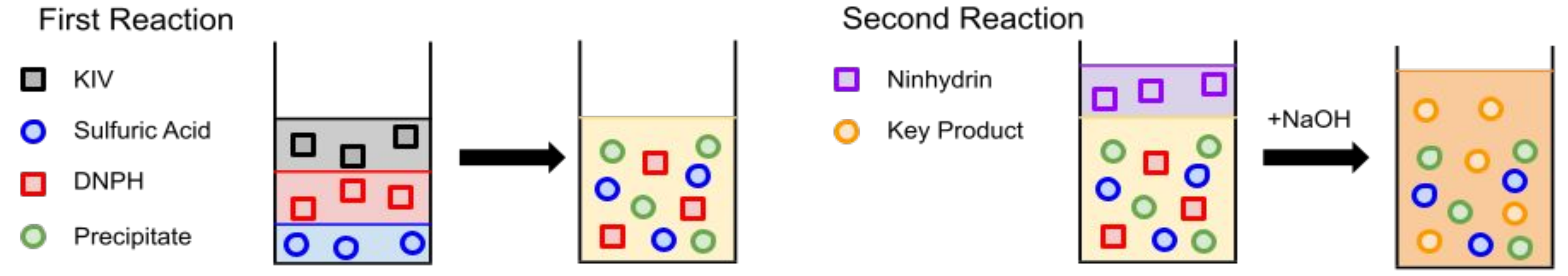
2,4-DNPH Test



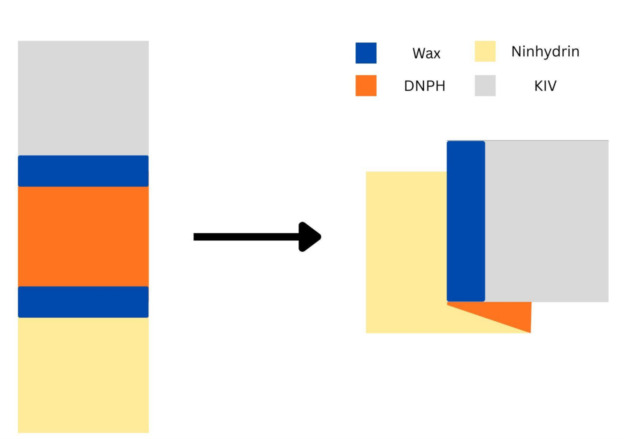
Leftover DNPH Detection[2]



In Solution:



On Paper:



1. Wax barrier is made between each section
2. Ninhydrin and DNPH is added to their respective sections
3. Empty section is folded over top the middle section and NaOH then KIV is added
4. Paper is wrapped in foil and heated

References

1. Hassan SA, Gupta V. Maple Syrup Urine Disease. [Updated 2024 Mar 3]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK557773/
2. Elgendy, K.; Alaa, M.; Elmosallamy, E.; Hassan, M.; Gamal, H. APPLICATIONS of NINHYDRIN: SPECTROPHOTOMETRIC DETERMINATION of HYDRAZINE and HYDROXYLAMINE (Aplikasi Ninhidrin: Penentuan Spektrofotometri Bagi Hidrazin Dan Hidroksil Amina). Malaysian Journal of Analytical Sciences 2022, 26, 855–866.
3. Structures drawn with Marvin JS by Chemaxon.

Results/Discussion

Figure 1. Spiking of 0.003 M KIV samples containing 10mM Brady's, 2% Ninhydrin, and 1x10^-6 M NaOH with KIV concentrations of 14.0 mM, 5.3 mM, 4.0 mM, 1.3 mM, 0.2 mM, and 0 mM.

- While lower concentration and tighter ranges of KIV concentrations could not yet be utilized for the origami folding mechanism, it is possible for the proper color changes to show on paper at low and tighter KIV concentrations.

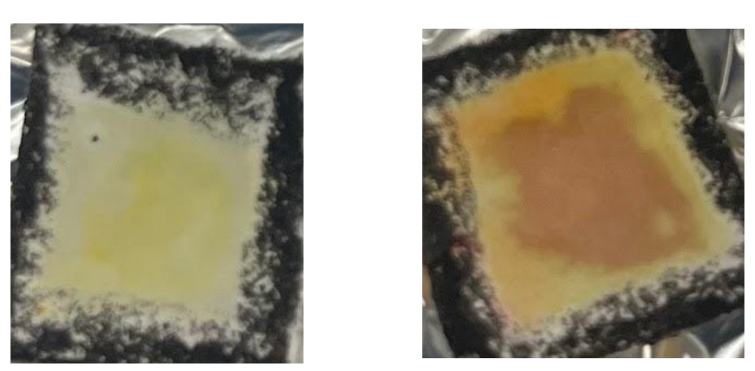


Figure 2. Proof of concept at low concentrations for paper tests. Whatman paper first soaked in 2% Ninhydrin, then has 5 mM Brady's, KIV (4 mM and 0.5 mM left to right), and 67 mM NaOH added in sequence with drying between each step.

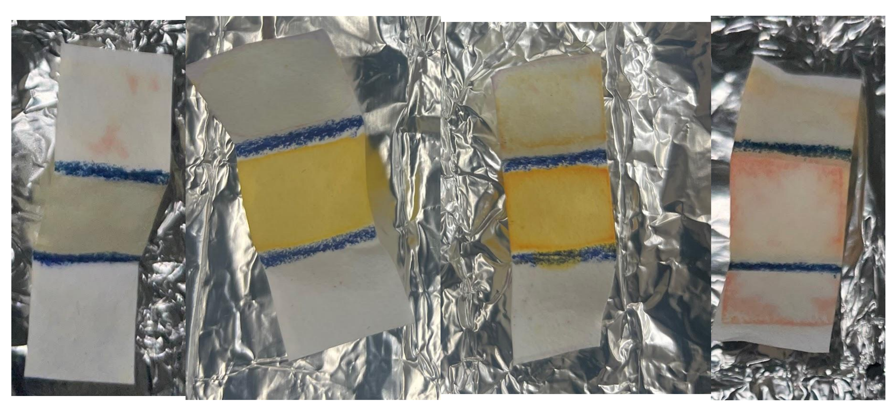


Figure 3. Paper tests using folding mechanism using 10mM Brady's, 2% Ninhydrin, and 1 μM NaOH with KIV concentrations of 143 mM (pos. control), 10 mM, 1mM, and 0 mM (neg. control).

- Positive and negative controls showed clear and correct differences in color, signifying that the reaction works properly using the mechanism.
- Spiking showed a lower limit of detection in solution, this limit has not yet been tested on paper

Next Steps

- Complete paperfluidic tests using folding mechanism at lower concentrations
- Show a greater color gradient with smaller gaps in concentration
- Develop a functional paperfluidic device and run test trials

Acknowledgements

Thank you to the First Year Innovation and Research Experience (FIRE) program at the University of Maryland.